Fatty acid synthesis

Objectives

- At the end of the lecture students should be able to:
- Describe the 3 steps involved in FA synthesis
- Describe the transport of acetyl CoA across mitochondrial membrane
- Describe the role of acyl carrier protein (ACP)
- Discuss the structure and function of fatty acid synthase

Fatty Acid Biosynthesis

Synthesis

- Cytosol
- Requires NADPH
- Acyl carrier protein
- D-isomer
- CO_2 activation
- Keto → saturated

Beta Oxidation

- Mitochondria
- NADH, $FADH_2$
- CoA
- L-isomer
- No CO_2
- Saturated → keto

Rule:

Fatty acid biosynthesis is a stepwise assembly of acetyl-CoA units (mostly as malonyl-CoA) ending with palmitate (C16 saturated)

3 Phases

Activation

Elongation

Termination

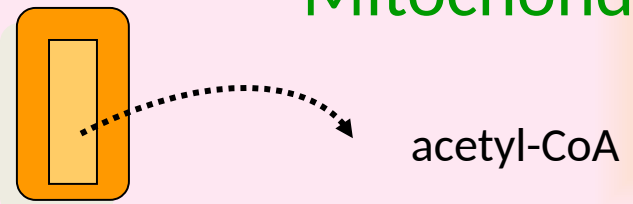
1. Transfer of acetyl-CoA from mitochondria to cytosol.

2. Activation of acetyl-CoA; synthesis of malonyl-CoA.

3. Five step elongation cycle of FA synthesis via ACP intermediates.

PROBLEM:

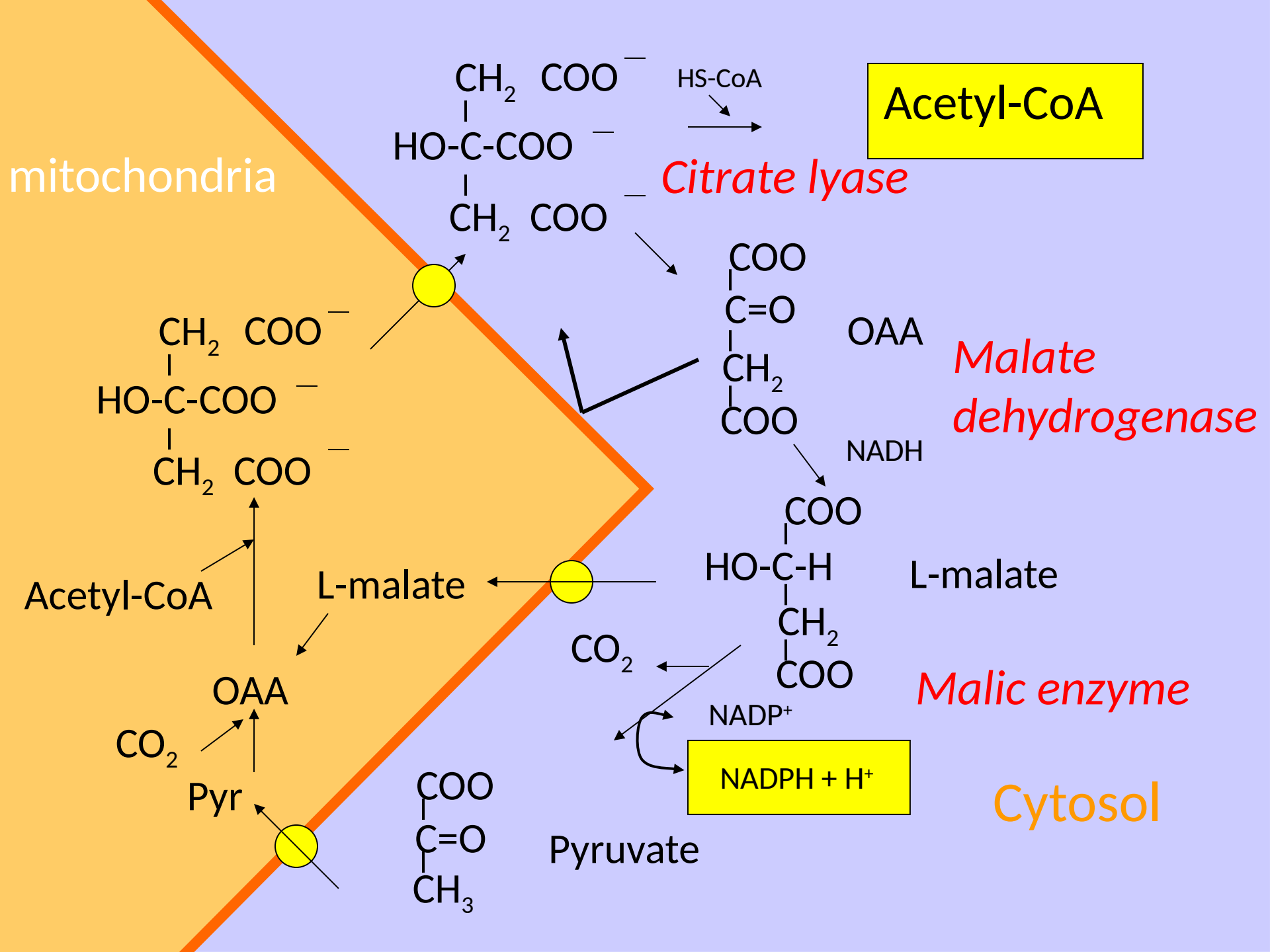
Fatty acid biosynthesis takes place in the cytosol. Acetyl-CoA is mainly in the Mitochondria



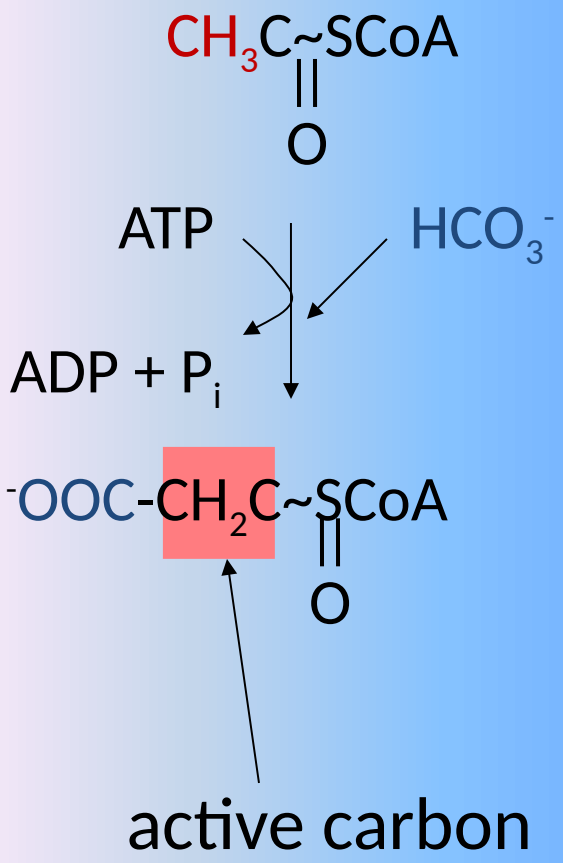
How is acetyl-CoA made available to the cytosolic fatty acyl synthase?

SOLUTION:

Acetyl-CoA is delivered to cytosol from the mitochondria as CITRATE

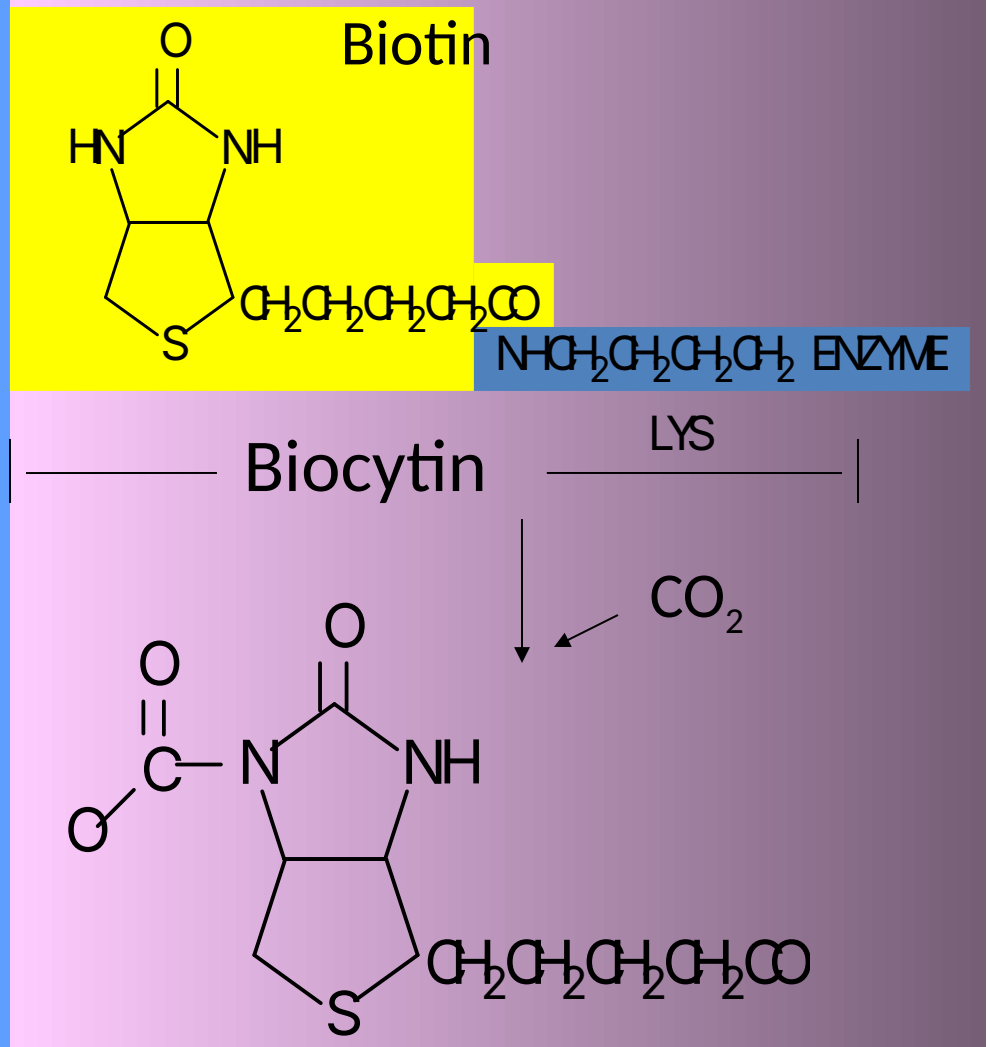

mitochondria
HS-CoA
Acetyl-CoA
Citrate lyase
OAA
Malate dehydrogenase
NADH
L-malate
L-malate
Acetyl-CoA
CO_2
Malic enzyme
OAA
$NADP^+$
CO_2
$NADPH + H^+$
Pyr
Pyruvate
Cytosol

ACTIVATION



Acetyl-CoA carboxylase

Cofactor



Carboxybiocytin

Formation of Malonylcoenzyme A is the committed step in fatty acid synthesis: It takes place in two steps: carboxylation of biotin (involving ATP) and transfer of the carboxyl to acetyl-CoA to form malonyl-CoA.

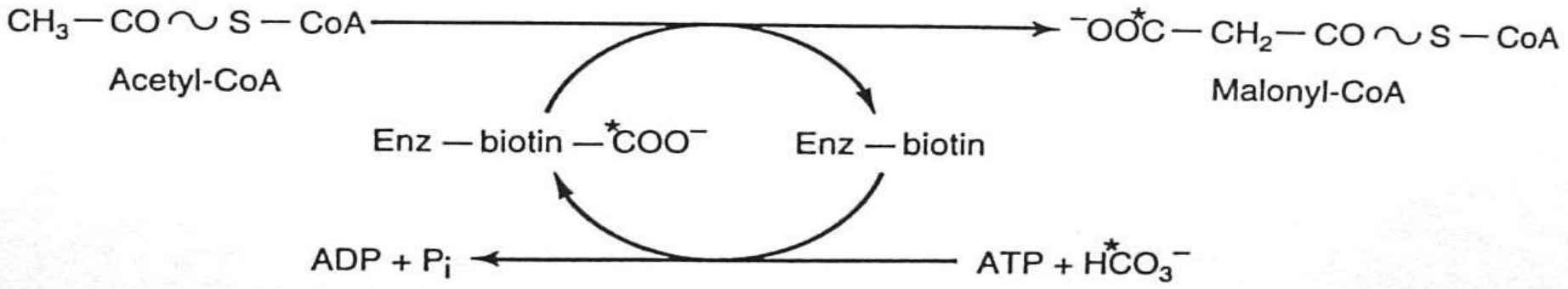


Figure 23–1. Biosynthesis of malonyl-CoA. (Enz, acetyl-CoA carboxylase.)

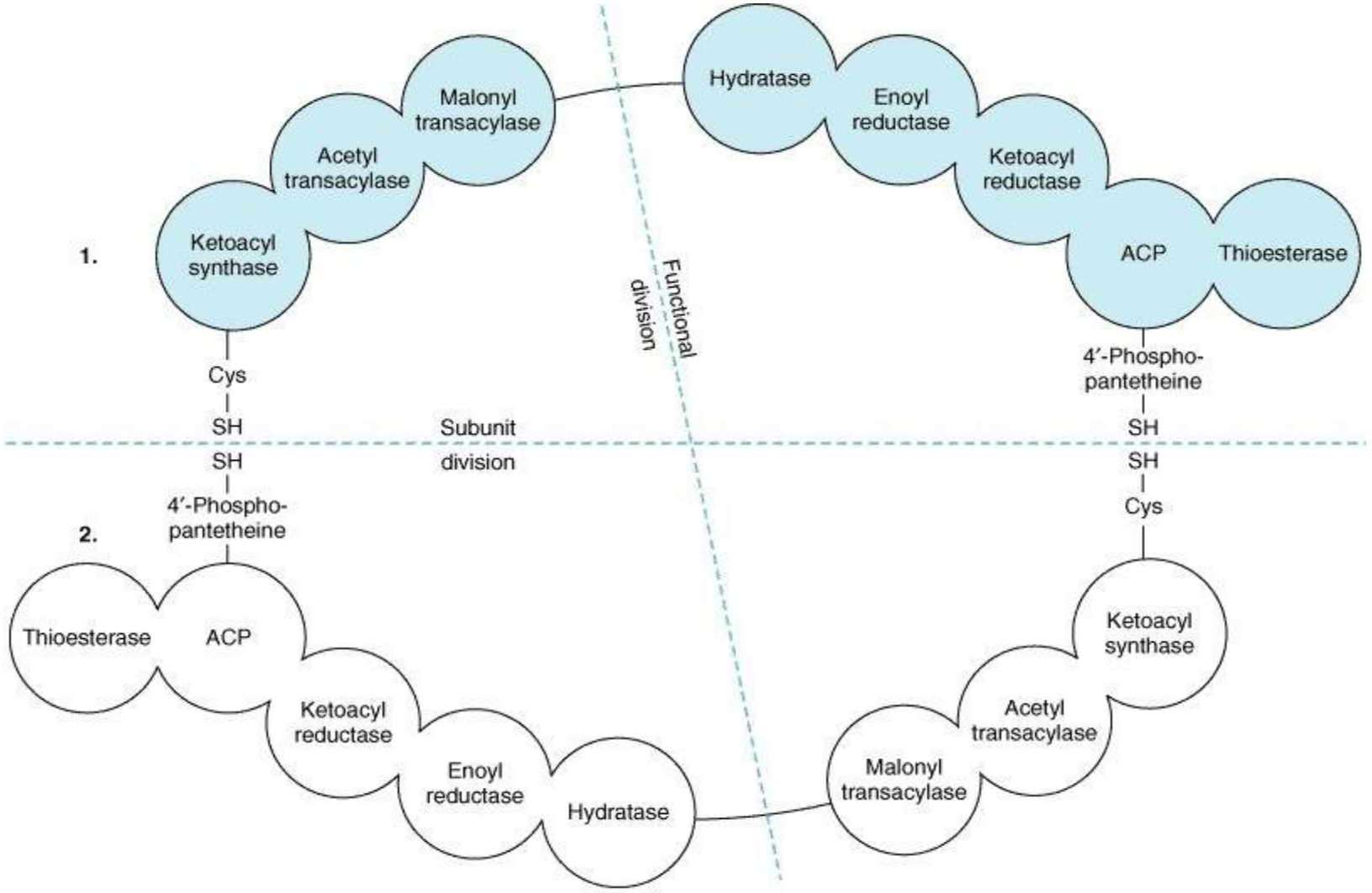
Reaction is catalyzed by acetyl-CoA carboxylase. It is a multienzyme protein. The enzyme contains a variable number of identical subunits, each containing biotin, biotin carboxylase, biotin carboxyl carrier protein, and transcarboxylase, as well as a regulatory allosteric site.

Fatty acid synthase

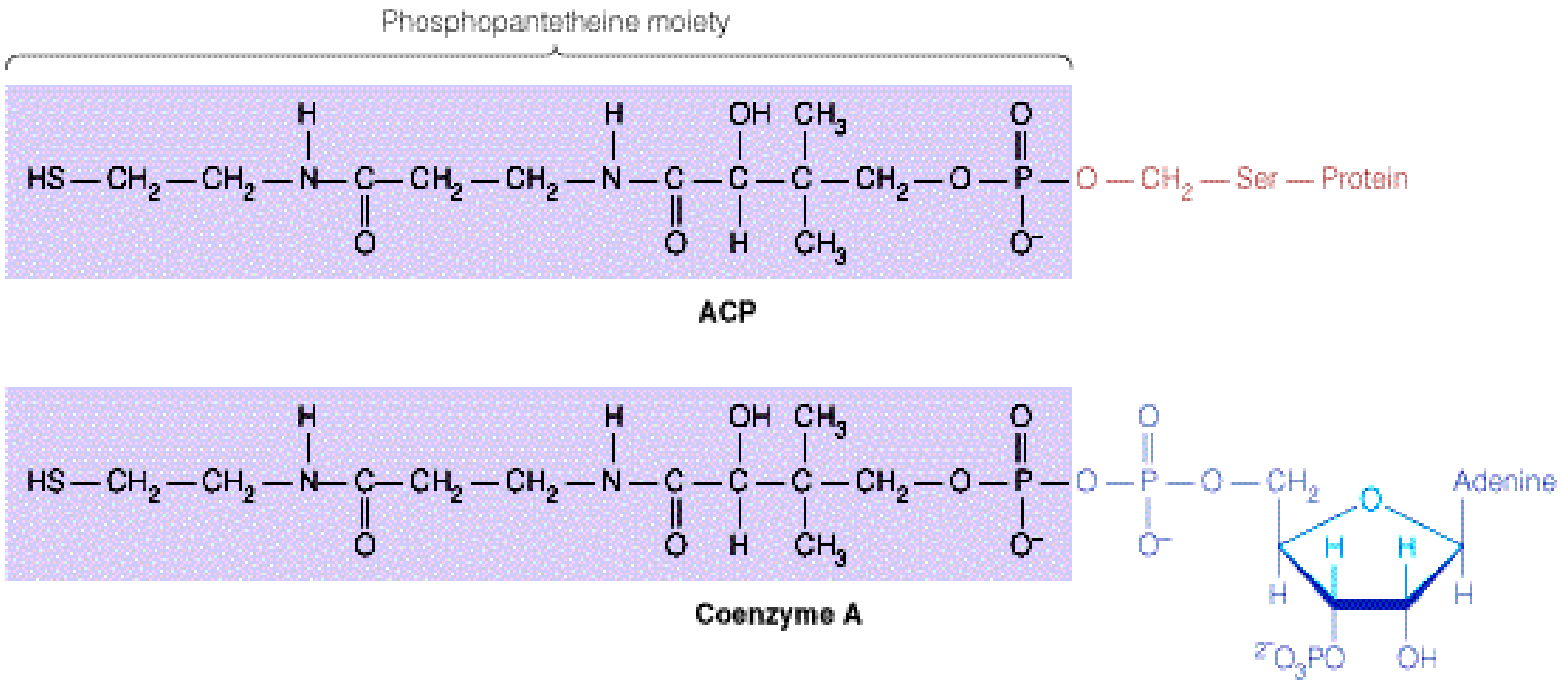
It is a multienzyme polypeptide complex that contains acyl carrier protein (ACP). ACP contains the vitamin pantothenic acid in the form of 4'-phospho-pantetheine. ACP takes over the role of CoA.

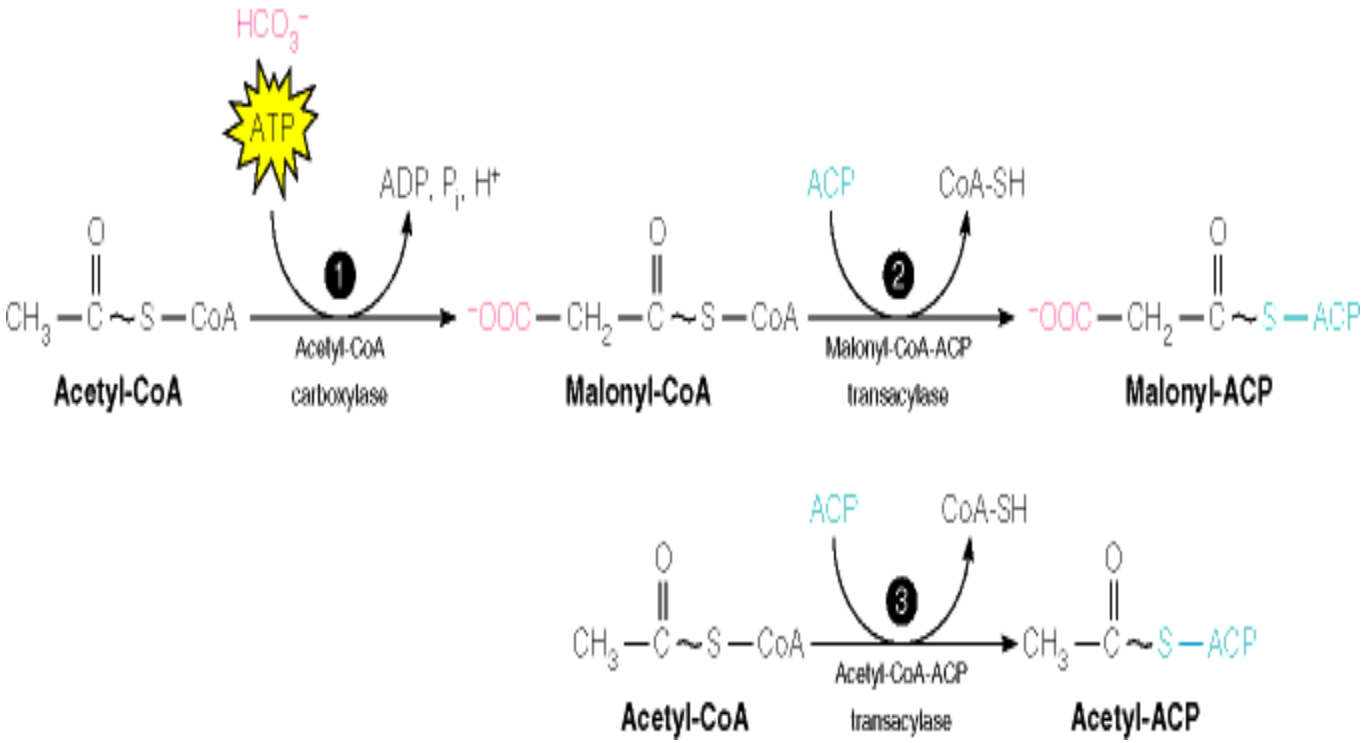
It is a dimer, and each monomer is identical, consisting of one chain containing all seven enzyme activities of fatty acid synthase and an ACP with a 4'-phosphopantetheine-SH group. Dimer is arranged in a "head to tail" configuration. Monomer is not active.

Fatty acid synthase

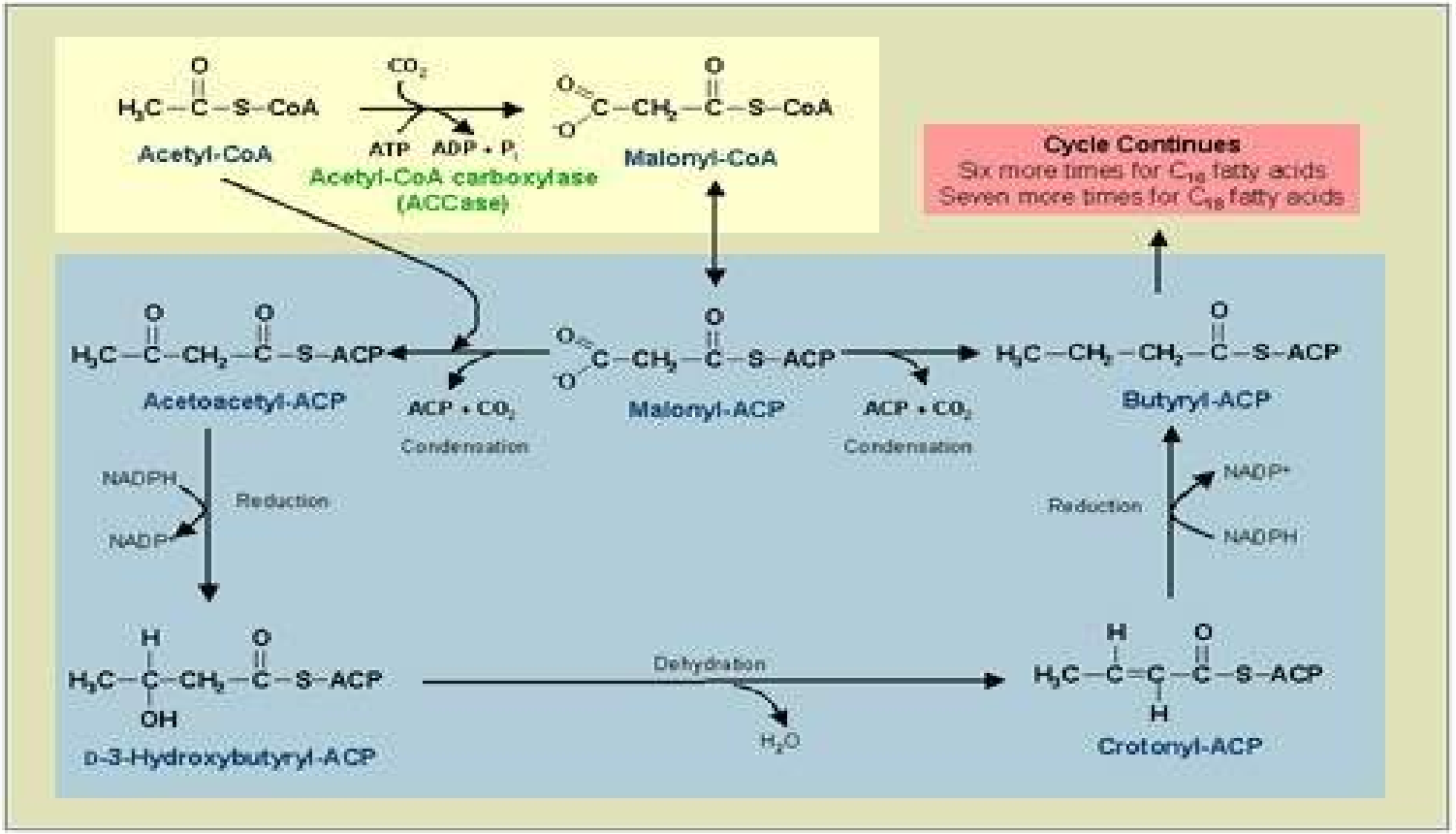


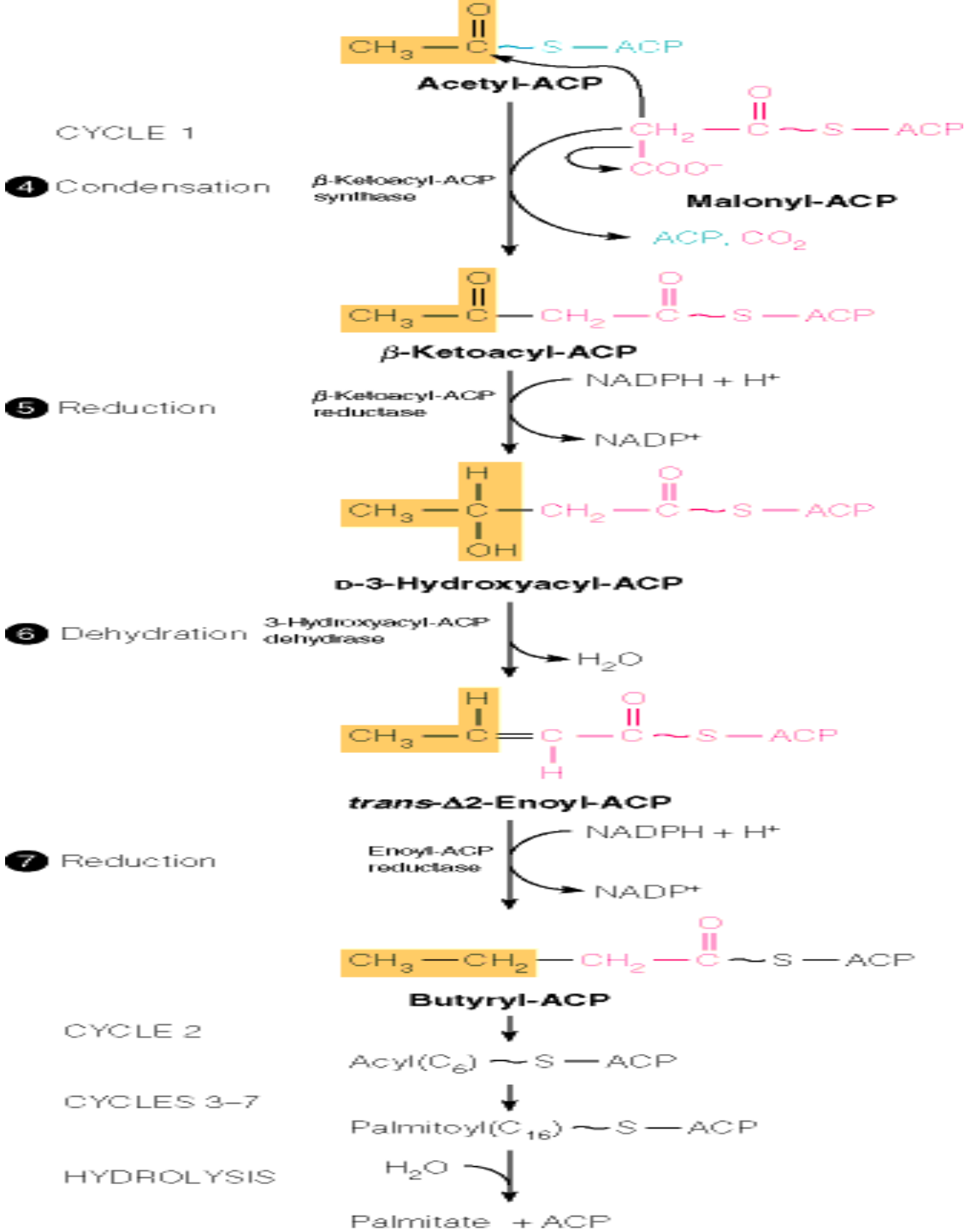
Coenzyme A and phosphopantetheine moiety ACP



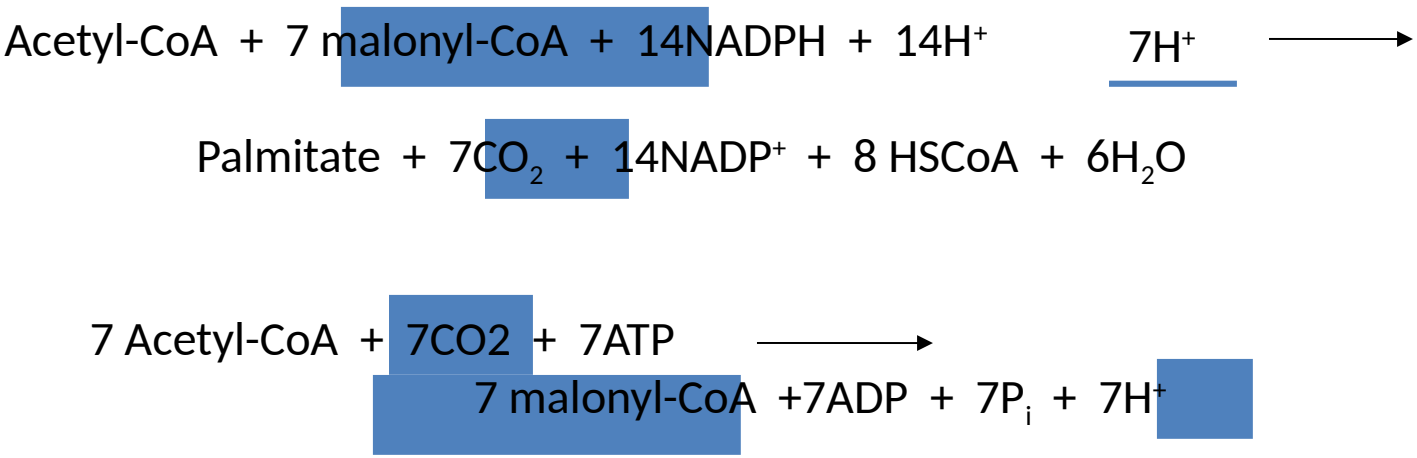

HCO3−
ATP
ADP, Pi, H+
ACP
CoA-SH
CH3—C~S—CoA
1
Acetyl-CoA carboxylase
Acetyl-CoA
-OOC—CH2—C~S—CoA
2
Malonyl-CoA-ACP transacylase
Malonyl-CoA
-OOC—CH2—C~S—ACP
Malonyl-ACP
ACP
CoA-SH
CH3—C~S—CoA
3
Acetyl-CoA-ACP transacylase
Acetyl-CoA
CH3—C~S—ACP
Acetyl-ACP

Fatty Acid Synthesis




CH3—C(=O)~S—ACP
Acetyl-ACP
CYCLE 1
4 Condensation
β-Ketoacyl-ACP synthase
CH2—C(=O)~S—ACP
COO⁻
Malonyl-ACP
ACP, CO2
CH3—C(=O)—CH2—C(=O)~S—ACP
β-Ketoacyl-ACP
5 Reduction
β-Ketoacyl-ACP reductase
NADPH + H+
NADP+
CH3—C(H)(OH)—CH2—C(=O)~S—ACP
D-3-Hydroxyacyl-ACP
6 Dehydration
3-Hydroxyacyl-ACP dehydrase
H2O
CH3—C(H)=C(H)—C(=O)~S—ACP
trans-Δ2-Enoyl-ACP
7 Reduction
Enoyl-ACP reductase
NADPH + H+
NADP+
CH3—CH2—CH2—C(=O)~S—ACP
Butyryl-ACP
CYCLE 2
Acyl(C6)~S—ACP
CYCLES 3–7
Palmitoyl(C16)~S—ACP
HYDROLYSIS
H2O
Palmitate + ACP

Overall Reactions



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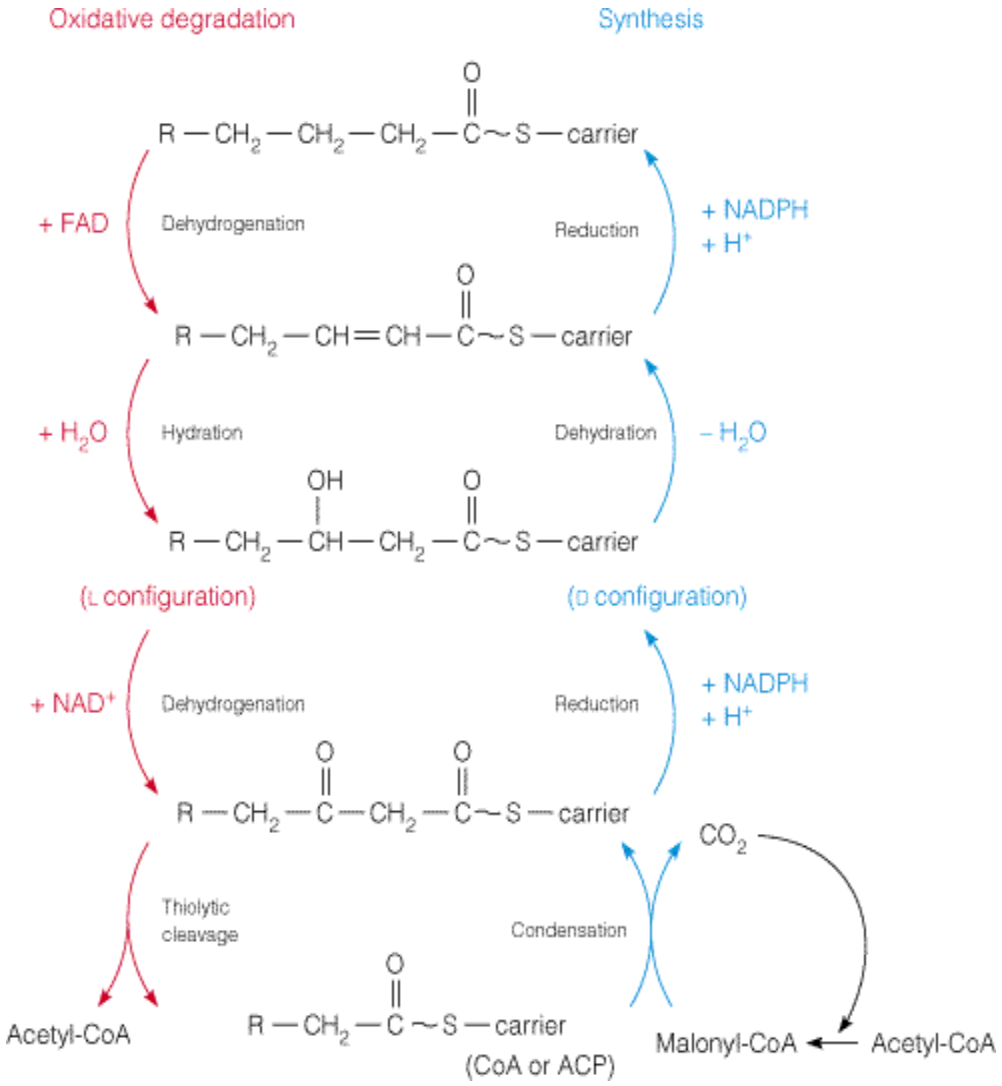
8 Acetyl-CoA + 14NADPH + $7H^+$ + 7ATP ⟶

Palmitate + 14$NADP^+$ + 8 HSCoA + $6H_2O$ + 7ADP + $7P_i$

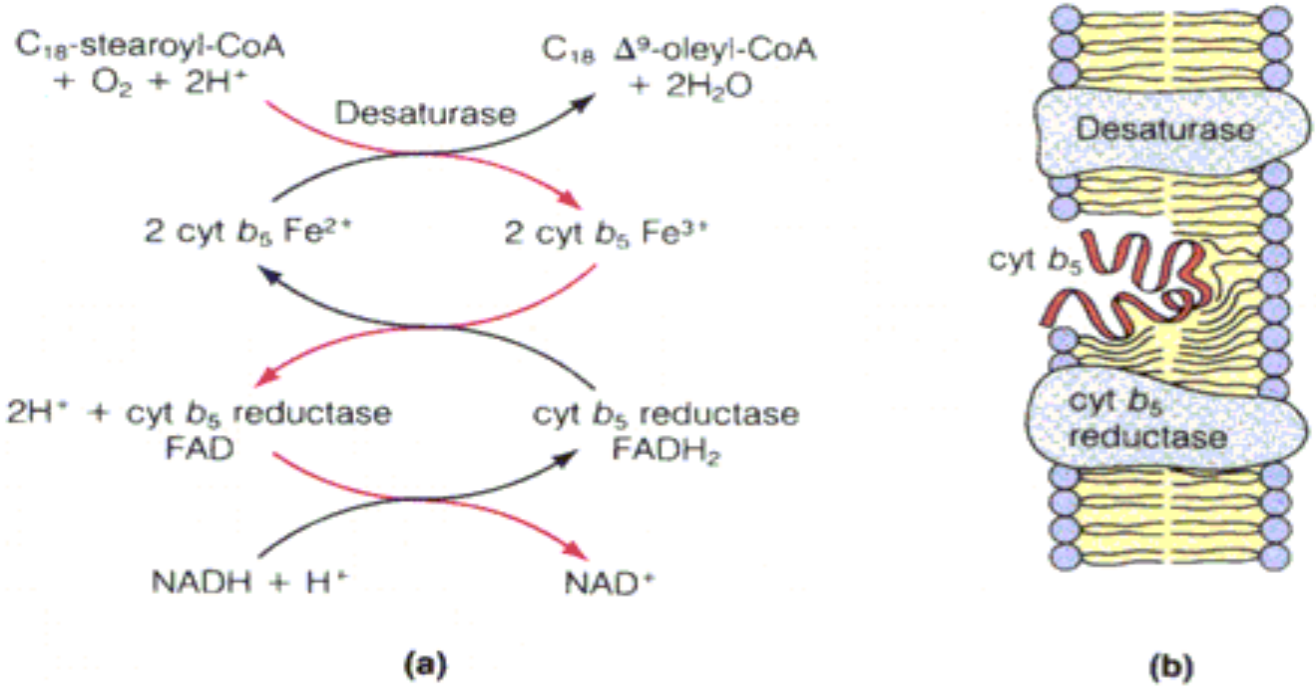
Enzymes

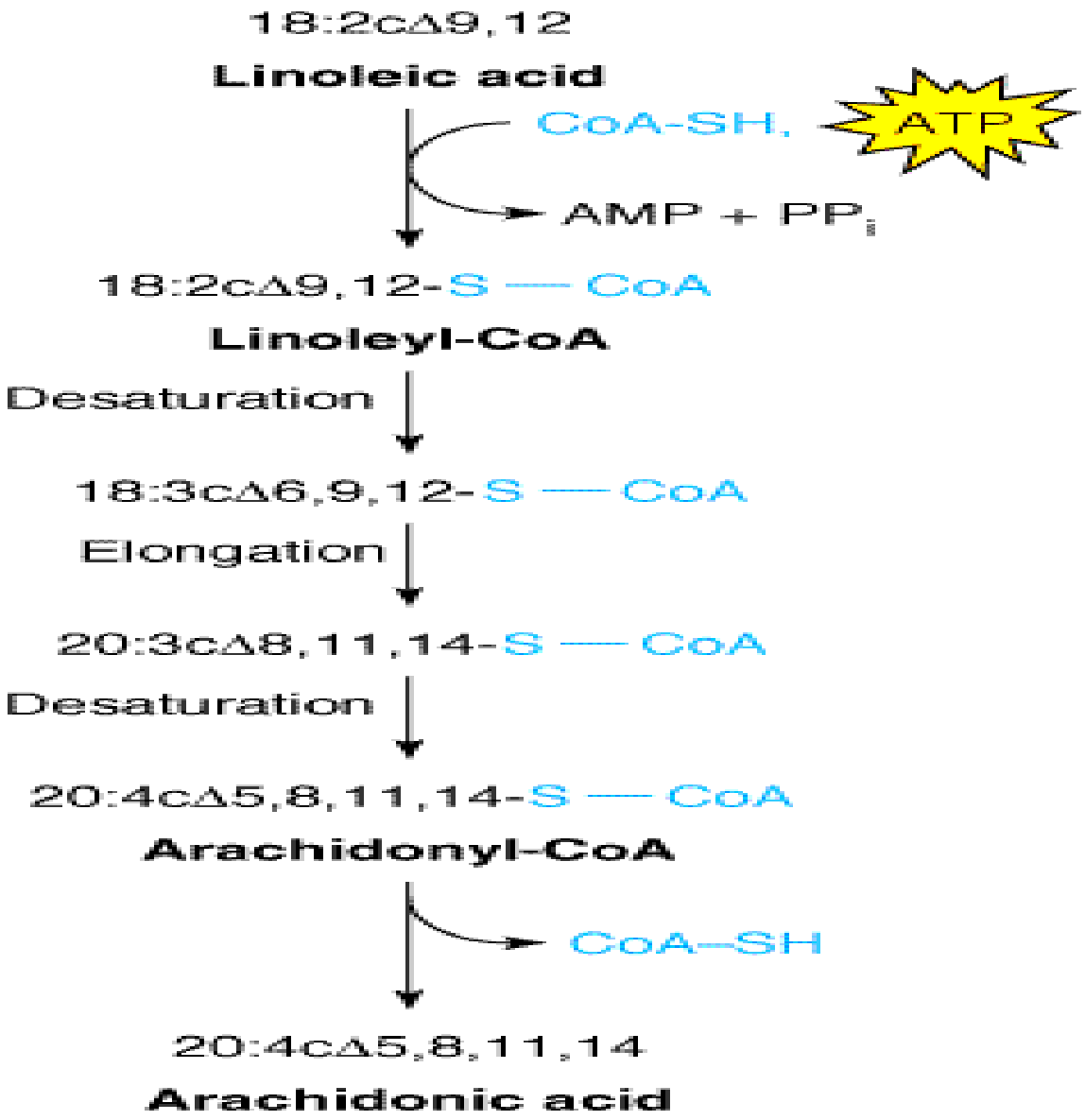
- AcetylCoA carboxylase
- Malonyl CoA- ACP transacylase
- Acetyl CoA- ACP transacylase
- b-ketoacyl ACP synthase
- b-ketoacyl ACP reductase
- 3-hydroxyacyl ACP dehydratase
- Enoyl ACP reductase

Fatty acid oxidation & synthesis



Desaturation




18:2cΔ9,12
Linoleic acid
CoA-SH,
ATP
AMP + PPi
18:2cΔ9,12-S — CoA
Linoleyl-CoA
Desaturation
18:3cΔ6,9,12-S — CoA
Elongation
20:3cΔ8,11,14-S — CoA
Desaturation
20:4cΔ5,8,11,14-S — CoA
Arachidonyl-CoA
CoA–SH
20:4cΔ5,8,11,14
Arachidonic acid

- Harper
- Lippincott
- Puri